Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

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Experiment 04

Verification of Thevenin's Theorem for a DC Network

Submitted By:
Raj Bhowmik
Reg. No: 202561901048
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Department of Software Engineering

Submitted To:
Md Rasedujjaman
Associate Professor
Department of Electrical and Electronic Engineering
Shahjalal University of Science and Technology

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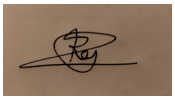
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Verification of Thevenin's Theorem for a DC Network"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit (EEE 0713 1206B)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

Sylhet, Bangladesh

Date: August 25, 2026



Raj Bhowmik
Reg. No: 202561901048

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Objective

- To understand the core principles of network reduction using Thevenin's theorem in linear, bilateral DC circuits.
- To construct a multi-resistor DC network and measure its open-circuit voltage (V_{th}) and equivalent resistance (R_{th}).
- To verify experimentally that the original network and its single-source Thevenin equivalent deliver identical current and voltage to a load.
- To evaluate agreement between empirical laboratory data and theoretical network calculations.

Introduction

Analyzing load response variations in complex networks becomes computationally repetitive if the complete circuit must be re-solved for every load modification. Thevenin's Theorem addresses this by modeling any two-terminal linear network—regardless of internal complexity—as a single equivalent voltage source connected in series with an equivalent resistance. This experiment demonstrates Thevenin's theorem by assembling a DC network, determining its equivalent parameters, and demonstrating terminal equivalence across a load resistor.

Theory

Thevenin's Theorem

Thevenin's Theorem states that any two-terminal, linear, bilateral DC network containing voltage sources and resistors can be represented by an equivalent circuit comprising an independent voltage source (V_{th}) in series with an equivalent resistor (R_{th}).

- **Thevenin Voltage (V_{th}):** The open-circuit voltage measured across terminal pair A–B when load R_L is removed.
- **Thevenin Resistance (R_{th}):** The equivalent resistance measured looking back into terminals A–B with all independent voltage sources replaced by short circuits (0 V).

Connected to load R_L, the load current I_L and terminal voltage V_L follow:

$$I_L = \frac{V_{th}}{R_{th} + R_L}, \qquad V_L = I_L R_L.$$

Relation to Norton's Theorem

Norton's Theorem offers a dual representation: replacing the network with an equivalent current source I_N in parallel with R_N, where $R_N = R_{th}$ and $V_{th} = I_N R_N$.

Apparatus and Components

Circuit Diagram

Original Circuit

Thevenin Equivalent Circuit

Table 1: List of Equipment and Components Used

Item Description	Specification / Rating	Quantity
DC Power Supply	15 V regulated	1
Resistor R_1	670 Ω	1
Resistor R_2	670 Ω	1
Resistor R_3	670 Ω	1
Load Resistor R_L	970 Ω	1
Digital Multimeter (DMM)	DC Voltage / Current / Ohms	1
Solderless Breadboard	Standard Tie-Point	1
Connecting Wires	Jumper Wires	As required

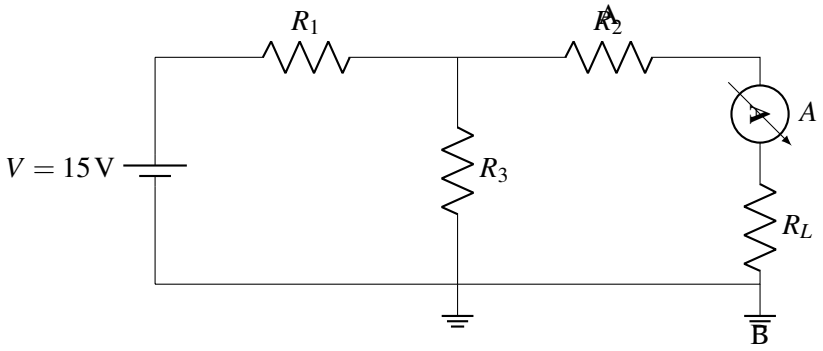


Figure 1: Original DC network featuring $R_1 = R_2 = R_3 = 670\,\Omega$ and load $R_L = 970\,\Omega$ driven by $V = 15\,\mathrm{V}$.

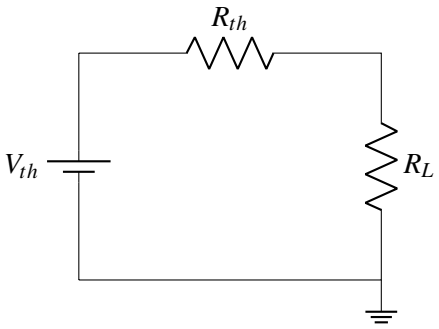


Figure 2: Thevenin equivalent circuit driving load R_L via series parameters V_{th} and R_{th}.

Experimental Procedure

Part A: Original Circuit Measurement

1. Assemble the primary circuit on the breadboard (Figure 1) with load R_L across terminals A–B.
2. Record load voltage (V_L) and current (I_L) using the digital multimeter.

Part B: Verification of Thevenin's Theorem

1. Disconnect load resistor R_L from terminals A–B.
2. Measure open-circuit voltage across terminals A–B to determine V_{th}.
3. Deactivate the DC source (short-circuiting supply terminals) and measure terminal resistance across A–B using an ohmmeter to determine R_{th}.

4. Build the equivalent model (Figure 2) on a separate breadboard section using V_{th} and R_{th}.
5. Connect load R_L across the equivalent circuit and record resulting load parameters.

Theoretical Calculation

With load R_L removed, open-circuit output matches the voltage division across R_3:

$$V_{th} = V\left(\frac{R_3}{R_1 + R_3}\right).$$

With source V short-circuited, equivalent input resistance equals $R_1 \parallel R_3$ in series with R_2:

$$R_{th} = R_2 + \left(\frac{R_1 R_3}{R_1 + R_3}\right).$$

Theoretical load parameters are computed as:

$$I_L = \frac{V_{th}}{R_{th} + R_L}, \qquad V_L = I_L R_L.$$

Data Collection

Table 2: Experimental Data for the Verification of Thevenin's Theorem

Parameter	Original Circuit (measured)	Thevenin Equiv. (measured)	Theoretical Value	% Error
Open-Circuit Voltage, V_{th}	N/A	8.23 V	8.105 V	1.54%
Equivalent Resistance, R_{th}	N/A	1.03 kΩ	978 Ω	5.3%
Load Current, I_L	4 mA	4.02 mA	4.16 mA	3.37%
Load Voltage, V_L	4 V	4.07 V	4.0352 V	0.86%

Result Analysis

Measured parameters ($V_{th} = 8.23\,\text{V}$, $R_{th} = 1.03\,\text{k}\Omega$) showed close agreement with theoretical values (8.105 V, 978 Ω), yielding minor percentage errors of 1.54% and 5.3%. Original load metrics ($I_L = 4\,\text{mA}$, $V_L = 4\,\text{V}$) closely matched parameters recorded from the physical equivalent circuit ($I_L = 4.02\,\text{mA}$, $V_L = 4.07\,\text{V}$), confirming electrical equivalence.

Discussion and Sources of Error

Experimental deviations are mainly attributed to:

1. **Resistor Tolerances:** Variance between coded and real resistance values influences both R_{th} measurements and theoretical outputs.
2. **Contact Resistance:** Contact resistance within breadboard strips and shorting links slightly shifts ohmmeter resistance readings.
3. **Multimeter Burden Voltage:** Ammeter internal resistance introduces minor loading effects during branch current readings.
4. **Equivalent Parameter Substitution:** Practical component tolerances prevent exact matching of calculated R_{th} on physical breadboards.

Conclusion

The experimental results validate Thevenin's Theorem within small operational error limits. The physical equivalent network replicated terminal behavior effectively, proving that a complex linear network can be reduced to a single source with series resistance.

References

1. Alexander, C. K., & Sadiku, M. N. O. (2016). *Fundamentals of Electric Circuits* (6th ed.). McGraw-Hill Education.
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